Saad Subhani

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Education

National University of Science & Technology (NUST) 2020 – 2024
Bachelor of Computer Sciences | CGPA: 3.63 *Islamabad, Pakistan*

Skills

Programming: C/C++ · Python · Java · MySQL · TypeScript · GraphQL · Intel MASM Assembly · Git/GitHub
Web: HTML/CSS/JavaScript/jQuery · MEAN/MERN Stack · React.js · NestJS · TypeORM
Machine Learning: PyTorch, scikit-learn, OpenCV, NLTK
DevOps: Microsoft Azure, Kubernetes, Docker, Keycloak, Redis, CI/CD, Nginx
Media: Photoshop, Illustrator, Premier Pro, Figma, FL studio

Professional Experience

Zones, LLC Jul 2024 – Present
Software Engineer *Remote*

- **Full-Stack Development:** Architected and implemented RESTful and GraphQL APIs using NestJS and TypeORM for high-performance internal server communications across the Zones Digital Platform.
- **IAM Expertise:** Part of the Access Studio development team; managing login flows and introducing features like user impersonation and custom authentication themes using Keycloak.
- **Management & Product:** Managed the end-to-end development of "Enablement Studio" as Product Manager and UX Designer; delivered a complete working MVP within 2 sprints.
- **DevOps & Performance:** Automated CI/CD pipelines to update container image tags on AKS clusters and utilized Redis Caching to optimize URL rendering and system stability.
- **Core UI Kit:** Developed and contributed core UI components for the ZDP toolkit using React and TypeScript; resolved critical bugs and performed manual QA across key modules like File System and Data Objects.

WALEE Nov 2022 – Oct 2023
MEAN and MERN Stack Intern *Islamabad, Pakistan*

- Developed full-stack web applications using the MEAN Stack, including designing, and implementing user interfaces, creating RESTful APIs, and integrating with databases including MongoDB and MongoDB Atlas.
- Collaborated with development teams to deliver high-quality software products, working in Agile environments, and using tools like Git for version control.
- Participated in project planning and management, contributing to requirements gathering, estimating development effort, and providing regular progress updates.

NCIL Nust Coventry Internet of Things Lab Sep 2022 – Nov 2022
Web Developer *Islamabad, Pakistan*

- Improved the user interface and functionality of the IoT Lab website, adding the ability for clients to communicate via email directly from the site.
- Demonstrated excellent communication skills, both written and verbal, with the ability to present technical information to non-technical stakeholders and collaborate effectively with cross-functional teams.
- Work done shall be reflected on the website: https://iotlab.seecs.nust.edu.pk/

Projects

X-rAI (Final Year Project) | *Winner of ICESCO Excellence Award*

- Leveraged the state-of-the-art ChexNet model to extract image embeddings from chest X-rays as part of the X-rAI project, enhancing our understanding of diagnostic imaging.
- Finetuned an advanced NLP model to efficiently generate comprehensive diagnostic reports from image embeddings, enhancing AI's role in medical decision making.

Olivia: A Multi-Functional Voice Assistant

- Engineered "Olivia," an intricate Windows voice assistant, integrating PyAutoGUI, OpenAI, Pygame, SpeechRecognition, and NLP to enable advanced capabilities.

- This encompassed traditional voice assistant features along with interactive voice-controlled games, highlighting a multidisciplinary technical approach and user-centered design.

S.H.A Share

- Led the development of S.H.A. Share, a secure, user-friendly distributed file-sharing system, emphasizing seamless file exchange across devices and locations while prioritizing data integrity and privacy.
- Hosted the S.H.A. Share website, enabling public access for widespread utilization of the secure file-sharing platform.

Obesity Management System

- Designed and developed an Obesity Management System as part of a research collaboration with Prof. Sungyoung Lee of Kyung Hee University, using React.js, Node.js, PHP, and MySQL.
- Implemented an MVC architecture to ensure clean separation of concerns and scalability and utilized HTML and CSS to create a responsive and user-friendly interface.

Hypertextual Web Search Engine

- Designed and developed a hypertextual web search engine based on the original research paper of Google, using Python and the Tkinter library for GUI.
- Preprocessed a large dataset of news articles and created data structures like a lexicon, forward index, and inverted index to efficiently store and retrieve information.

PONG in Assembly

- Developed a two-player arcade game of PONG using Intel MASM Assembly architecture, demonstrating a strong understanding of low-level programming concepts and computer hardware.
- Utilized Assembly language to optimize performance and reduce memory usage, implementing game mechanics such as collision detection, player movement, and score tracking.

Volunteer Work & Community Service

Pehli Kiran Schools Islamabad, Pakistan

Industrial Tour Coordinator & Workshop Facilitator *2023*

- Organized and led an industrial tour for underprivileged children from Pehli Kiran School Numbers 5 and 8 to SEECS NUST; conducted educational sessions showcasing electrical engineering tools.
- Designed and delivered workshops on Canva, communication skills, freelancing, and portfolio creation; liaised with school administration to provide educational resources and develop students' soft skills.

NUST Student Societies: NUST Islamabad 2021 – 2024

Director & Deputy Director

- Director of Graphic Designing at YES (SEECS)
- Deputy Director Admin Events at NUST Digital Club